

Specifications

Laser Scanning Performance

Item	Cygnus 3	Cygnus 3 Pro
LiDAR Channels	16 channels	32 channels
Point Rate	320,000 pts/s	640,000 pts/s
Measurement Range	Up to 120 m	Up to 300 m
Field of View	360° × 280°	360° × 280°
Relative Accuracy	1 cm	1 cm
Point Cloud Thickness	1 cm	1 cm
Absolute Accuracy	Up to 4 cm*	Up to 2 cm*

Mapping & Positioning

Item	Specification
Mapping Modes	SLAM / RTK-SLAM / PPK-SLAM
Real-Time Processing	Yes
Real-Time Point Cloud Preview	Yes
Accuracy Heatmap	Yes
Resume Scanning	Yes
RTK Protocol	NTRIP

Camera System

Item	Specification
Resolution	2 × 48 MP + 1 × 5 MP
Camera FOV	190° × 190° for 48 MP; 135° × 100° for 5 MP

General Specifications

Item	Cygnus 3	Cygnus 3 Pro
Weight	1.98 kg	1.68 kg
Battery	70 Wh	70 Wh
Operating Time	Up to 180 min	Up to 180 min
Storage	512 GB SSD	512 GB SSD
IP Rating	IP54	IP54
Operating Temp.	-20 °C to +55 °C	-20 °C to +55 °C
Interfaces	Type-C / SkyPort / GNSS	Type-C / SkyPort / GNSS

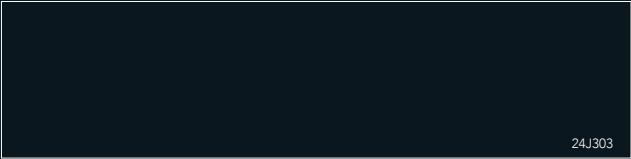
Notes Cygnus 3 is equipped with a 16-channel LiDAR delivering up to 320,000 pts/s. Cygnus 3 Pro features a 32-channel LiDAR with a point rate of up to 640,000 pts/s. Accuracy values are obtained under satlab standard calibration conditions. Performance may vary depending on environment, operating method, and GNSS availability. Battery endurance depends on operating temperature and working conditions.



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Cygnus 3

Series Scanner

Engineering-Grade SLAM Scanner

Integrated RTK-SLAM | Real-Time Quality Check | Multi-Platform Mapping

Integrated RTK

Accuracy Heatmap

Real-Time Point Cloud Preview

Handheld / Backpack / Vehicle / UAV



320k / 640k pts/s

Point Rate

120 m / 300 m

Measurement Range

RTK / PPK
GCPs + Resume
SLAM Workflow Support

2 × 48 MP
1 × 5 MP
Colorization &
Visual-Aided Positioning

Cygnus 3 series Scanner

Cygnus 3 is built for complex indoor, outdoor, and GNSS-challenged environments. It combines LiDAR, vision, RTK positioning, and advanced SLAM algorithms to deliver reliable 3D data for surveying, construction, infrastructure, and digital twin workflows.



Integrated RTK

Built-in RTK for globally consistent point cloud results.



Accuracy Heatmap

Check data quality directly during field scanning.



Multi-Platform Mapping

Handheld, backpack, vehicle, and UAV operation.



Rich Deliverables

Point clouds, mesh, 3DGS, and engineering outputs.



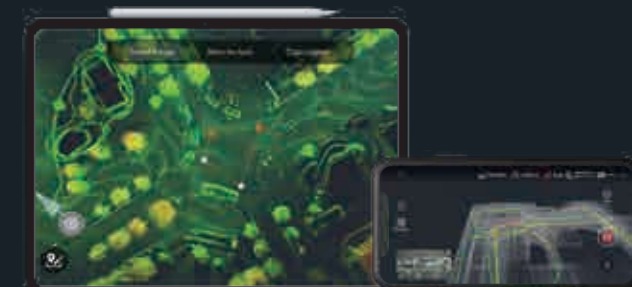
Key Features

Industry-Leading SLAM with Integrated RTK

- Advanced SLAM algorithms for complex environments
- RTK / PPK / GCP-supported workflows
- Improved coordinate consistency
- Clean setup with fewer external accessories

Real-Time Data Confidence

- Real-time point cloud preview
- Accuracy heatmap feedback
- Resume scanning support
- Verify data quality on site



Available on both Android and iOS platforms

Multiple Output Forms

- High-Accuracy Point Clouds ---Engineering-grade geometry for surveying and analysis
- Mesh Surface Models ---Structured surfaces for inspection, visualization, and measurements
- 3DGS Reconstruction & Visualization ---Visual representations for presentation, communication, and digital twins

Flexible Operation Modes



- Handheld Mode
- Backpack Mode
- Vehicle-Mounted Mode
- UAV-Mounted Mode



SAT-LIDAR SOFTWARE WORKFLOW

From SLAM Data to Engineering Deliverables



Cygnus 3 works with Sat-LiDAR to connect field scanning, SLAM processing, correction, analysis, visualization, and deliverable preparation in one workflow. Users can move from raw scan data to engineering-ready outputs without breaking the workflow.

Powered by Sat-LiDAR

A connected workflow for SLAM processing, correction, analysis, visualization, and delivery.

Sat-LiDAR Software Workflow & Ecosystem

1

Import

Cygnus 3 raw scan data

2

Process

Optimize, correct and georeference

3

Analyze

Measure, section and calculate

4

Deliver

Generate final deliverables

Process & Correct

Convert Cygnus 3 raw scan data into optimized, georeferenced, and colorized point clouds.

- SLAM processing and optimization
- Colorized point cloud generation
- Coordinate transformation and georeferencing
- RTK / PPK / GCP-supported correction
- Accuracy check and data adjustment

Analyze & Deliver

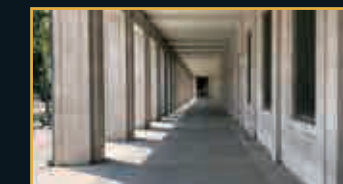
Turn processed data into engineering, visualization, and digital twin deliverables.

- DEM, contours, sections, and profiles
- Volume measurement and engineering analysis
- Mesh and 3DGS visualization
- CAD / BIM reference data output

Application Scenarios



Urban Mapping



Infrastructure & Corridors



Industrial Facilities



Architecture
& Construction



Digital Twins
& Visualization



Emergency Response
& Rapid Mapping